

Generation Interconnection Request System Impact Study Report

For Queue Project Q21-1724

Idle Hour 100 kV
August 22, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1724. The Interconnection Customer proposes a 132.2 MW solar plus storage facility interconnecting at the Idle Hour 100 kV substation in the Michigan area. The request was received on April 18, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 527 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$320,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The DUNLAP - SNEATH-PACIFICA TAP 100 kV is overloaded to 104.7% of its 166 MVA emergency rating for the loss of the IDLE HOUR - STUBEY ROAD 100 kV. The project contributes 62.7% DFAX (82.9 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
DUNLAP - SNEATH-PACIFICA TAP 100 kV	Reconductor line	\$998,000	32% (\$320,000)
	Total allocated to project		\$320,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.